

DA5401: Data Challenge Report

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Problem Statement:

The goal was to predict the “Fitness Score (0-10)” of AI generated Response to a user prompt, evaluated against a specific metric definition. (Metric Definition Embeddings are provided)

EDA:

TRAIN - BASIC INFORMATION:

The train data shape-> 5000 rows × 5 columns

Column Types:

object 4

float64 1

Name: count, dtype: int64

Train Column Details:

Column	Type	Non-Null	Null	Null %	Unique	Unique %
metric_name	object	5000	0	0.00	145	2.90
score	float64	5000	0	0.00	12	0.24
user_prompt	object	5000	0	0.00	4886	97.72
response	object	4999	1	0.02	4878	97.56
system_prompt	object	3451	1549	30.98	851	17.02

TEST- BASIC INFORMATION

Shape: 3638 rows × 4 columns

Column Types:

object 4

Name: count, dtype: int64

COLUMN DETAILS:

Column	Type	Non-Null	Null	Null %	Unique	Unique %
metric_name	object	3638	0	0.00	145	3.99
user_prompt	object	3638	0	0.00	3590	98.68
response	object	3637	1	0.03	3590	98.68
system_prompt	object	2532	1106	30.40	718	19.74

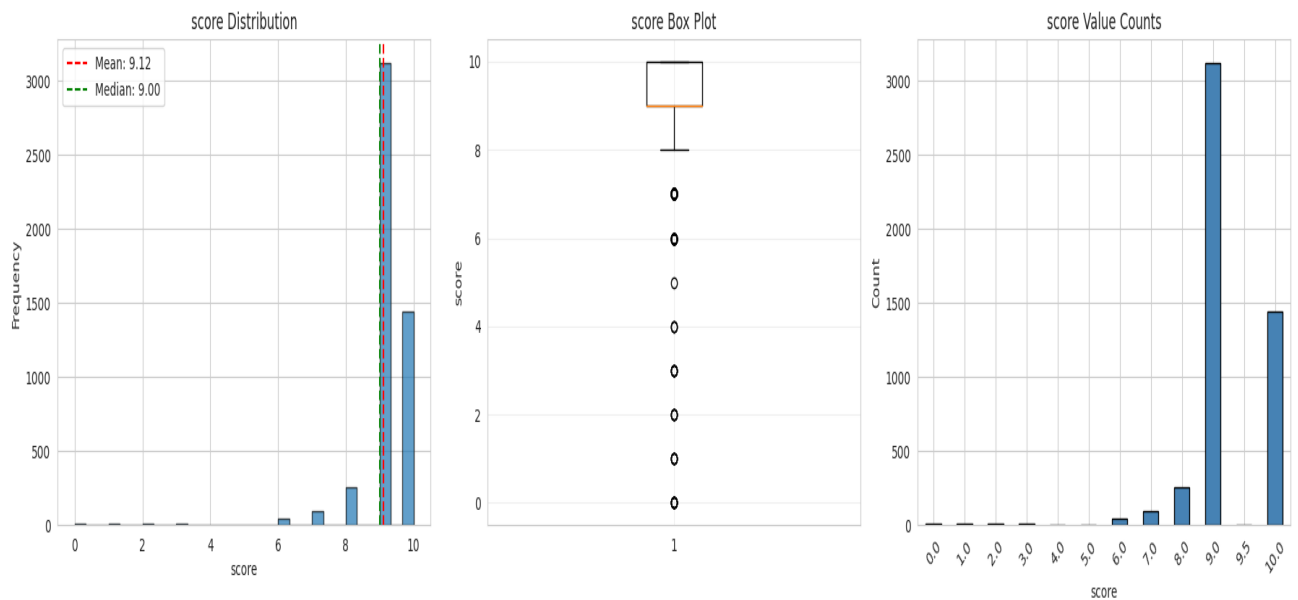
TARGET VARIABLE ANALYSIS: ('score')

Target Statistics:

Mean: 9.1195
Median: 9.0000
Std: 0.9424
Min: 0.0000
Max: 10.0000
Skewness: -4.4488
Kurtosis: 34.6916

Target Distribution:

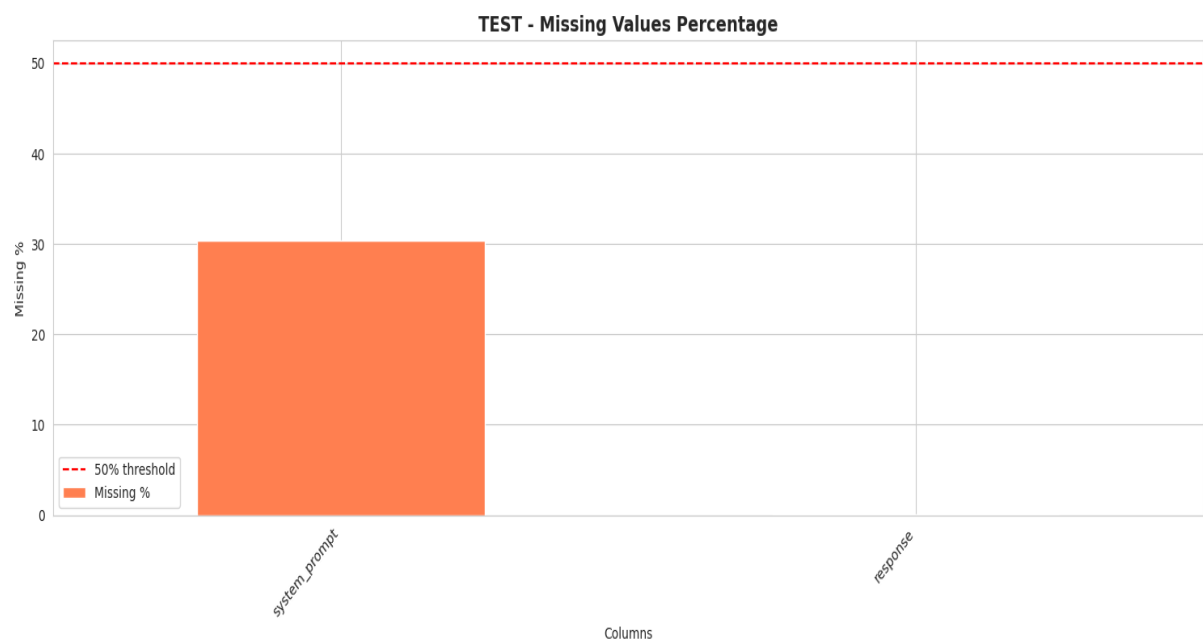
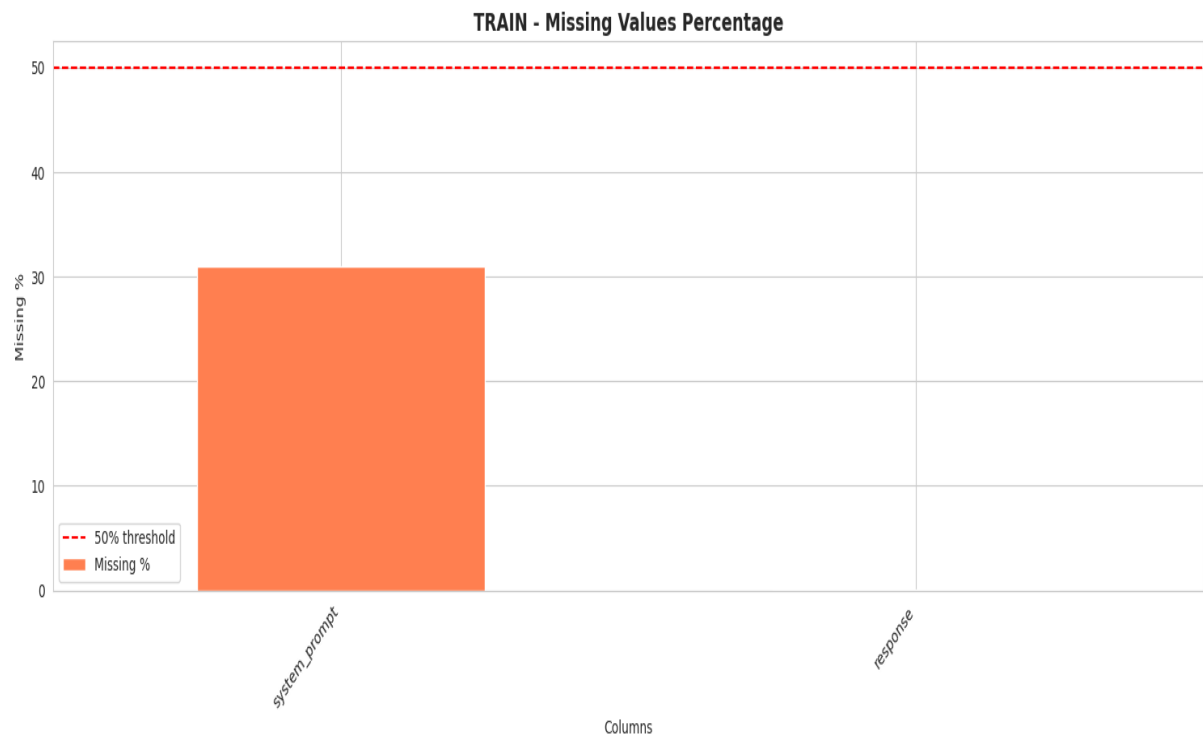
0.0: 13 (0.26%)
1.0: 6 (0.12%)
2.0: 5 (0.10%)
3.0: 7 (0.14%)
4.0: 3 (0.06%)
5.0: 1 (0.02%)
6.0: 45 (0.90%)
7.0: 95 (1.90%)
8.0: 259 (5.18%)
9.0: 3123 (62.46%)
9.5: 1 (0.02%)
10.0: 1442 (28.84%)



Class Imbalance Ratio:

Majority/Minority: 3123.00:1
(This showcases the extreme train data skewness)

TRAIN and TEST MISSING VALUES ANALYSIS:



Conclusion:

1. SKEWED FEATURES

2. HIGH CARDINALITY CATEGORICAL:

- metric_name: 145 unique values
- user_prompt: 4886 unique values
- response: 4878 unique values
- system_prompt: 851 unique values

Data Engineering and Feature Strategy:

1. Data Engineering:

I used two types of embedding, combining generalist sentence embeddings with specialist regional embeddings to capture the full semantic representation of the prompt-response pairs.

- **Dual Embedding Space:** I used LaBSE (Language-Agnostic BERT) and IndicBERT (L3Cube) embeddings (Both provide 768x1 dimension embedding). Because LaBSE provides robust understanding for English and General concepts, whereas IndicNLP captures the cultural and linguistic nuance of the native Indian languages
- **Feature Factory:** I engineered 56 geometric features to represent the relationship between prompt(P) – response(R) and metric(M).
 - **Cosine Similarity:** Cosine similarity between (R,M), Cosine similarity between (P,M), Cosine similarity between (P,R).
 - **Euclidean Distances:** To capture the magnitude the differences that the cosine similarity ignores.
 - **Scalar Projections:** To measure how much the “Response Vector” lies in the direction of the “Metric Vector”.
 - **Polynomial Interactions:** I created this polynomial feature like eg. $\text{Cosine Sim}(R,M) * \text{Cosine Sim}(P,M)$, in order to help the tree based models to learn that high relevance only matters if the task difficulty is significant.
- **System Context via Kernel PCA:** I processed the “System Prompts” using Nystroem Kernel PCA with 95% variance retention. This can capture the non-linear “Persona” clusters without dimensions of the features.

2. Sampling Strategy:

The training dataset is extremely skew, hence this part is very important.

- **Training Distribution:** Had mean score 9.12 and 91% data had score greater than or equal to 8.0
- **Test Distribution:** Followed completely different score distribution (mean was close to ~6 or ~7, concluded from the means of the predictions which were getting better test rmse).

Hence, I followed “**Negative Sampling**” method to augment the data, assuming that the prompt – response pair which are getting high score for a given metric name might have lesser score for the other metric name.

Methodology:

- Identified high score prompt- response pairs from the training dataset.
- Paired them with the Dissimilar Metrics (whose cosine similarity was < 0.4) from the global pool.
- Assigned these new datapoints less score in the range of [1.0, 2.0]
- Generated 2000 synthetic negative samples.

Hence as a result the regressors and classifiers were not getting biased to predict the high scores always.

3. Further Insights:

- Cluster Analysis: I used kmeans and identified 5 distinct semantic domains, may be representing subsets/regions like Safety related, Factualness related or Creativity Related, etc. This gave me the idea of using “**Mixture of Experts**”.
- Anomaly Detection: I used Isolation Forest and found that the valid low scores in the training dataset were often the geometric outliers.

Model Selection (Heterogeneous Mixture of Experts):

I divided the architecture to global and local committees (ensemble of experts), where each local committee was trained on the particular cluster of datapoints. (5 committees of experts)

Each of the “Committee” (Heterogenous Ensemble) has 4 different models to minimize the variance:

1. LightGBM Regressor: Optimized for MAE
2. XGBoost Regressor: Optimized for Absolute Error
3. CatBoost Regressor: Excellent for noisy data.
4. LightGBM Classifier: To capture the probability distribution.

Also I used soft gating, a “Gate” weighted the contribution from each expert based on the sample’s domain from the centroid of the domain, using the softmax function. Hence the final prediction is weighted average of the 5 experts based on how each domain is relevant to the specific sample.

Finally, I used meta learner (stacking), collected the Out of Fold (OOF) predictions from all 5 experts and trained a Ridge Regression meta learner. This helped the model to learn global biases and correct for any errors in the individual experts.

Performances of Different approaches:

1. **Baseline LGBM:** Trained the simple standard LGBM regression using raw LaBSE embeddings. The model achieved very low oof rmse and the model basically was an overfit.
2. **Stacked Ensemble:** I considered training an ensemble of lightGBM, XGBoost and CatBoost using Ridge regressor as meta learner.
3. **Heterogenous MoE:** I used the domain based Mixture of Experts model as explained earlier in Model Selection, this resulted in test rmse of 3.02. This architecture was capable of capturing the complex and non linear relationship.
4. **Mixture of Ensembles(MoE) and Transductive Augmentation :** Since the train and test distributions were different, in order to bridge the distribution gap, I attempted to generate some synthetic negative samples using **Test set prompts** paired with mismatched metrics. But the approach failed may be it confused the model during inference.
5. **Heterogenous MoE with Optuna Tuning (Final Best Model):** Fine tuned the hyperparameter in third approach using optuna. Gave extra reduction in test rmse.

Experiment / Approach	OOF RMSE (Original Data)	Test RMSE	Key Observation
Baseline (Single LGBM)	0.512	4.136	The model was overfit and biased to train data, with mean (9.1) of predicted scores. Failed on Test.
Stacked Ensemble	2.570	3.063	Good architecture, but lacked negative samples.
Hetero MoE	2.133	3.020	Strong performance. With MoE helped
MoE + Transductive	1.822	3.175	Failed. Test-side augmentation introduced noise.
MoE + Optuna Fine Tuning	1.915	2.968	Best Result, with finetuning